

Highlights

Conditional Diffusion-based Sampling Method for Surrogate-based Aerodynamic Shape Optimization

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- Conditional diffusion-based sampling method
- Generative method for high-fidelity high-performance wing samples
- Generated samples support surrogate-based aerodynamic shape optimization with lower computational cost

Conditional Diffusion-based Sampling Method for Surrogate-based Aerodynamic Shape Optimization

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Abstract

The development of machine learning methods has greatly advanced the resolution of surrogate-based design optimization problems. However, these approaches are still constrained by the trade-off between the need for large volumes of data and the dimensionality of the design parameters. Specifically, in high-fidelity design problems, handling high-dimensional geometric parameterizations requires large datasets to ensure that the surrogate model achieves the necessary accuracy for reliable aerodynamic performance estimation. This accuracy is crucial for providing good estimates of state variables and sensitivity analysis in subsequent gradient-based optimization. However, the process of generating such large datasets is computationally inefficient from a cost perspective. The strategy of trading substantial computational cost for data to solve design optimization remains a point of contention. To address this issue, we propose a deep generative learning model that leverages historical data from gradient-based optimization. By relying on a generative model—specifically, a conditional diffusion model—we aim to harness its capability to model high-dimensional data spaces and achieve inverse design by imposing relevant aerodynamic requirements on the model.

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1. Introduction

Surrogate-based aerodynamic shape optimization (ASO) has significantly transformed the landscape of aircraft design by effectively balancing computational efficiency and optimization accuracy. By substituting computationally expensive computational fluid dynamics (CFD) simulations with surrogate models, this approach enables rapid performance predictions and facilitates extensive exploration of the aerodynamic design space. Recent studies have demonstrated the effectiveness of surrogate-based approaches in addressing high-dimensional optimization [1], integrating multifidelity information [2], and operating within trust-region-constrained frameworks [3] for aerodynamic shape optimization.

During the pursuit of surrogate-based ASO, a common dilemma arises when striving for high-fidelity, high-dimensional optimization. On the one hand, a large number of aerodynamic data are required for training to ensure accurate surrogate model constructions. Recent research by Li and Zhang [4] has validated that, by leveraging sufficiently large datasets, surrogate-based methods can achieve optimization results that are competitive with those obtained from state-of-the-art adjoint-based CFD solvers. However, the reliance on traditional Latin Hypercube Sampling (LHS) for generating training samples still presents certain limitations. Firstly, these space-filling sampling methods aim to ensure uniform coverage of the design space but often fail to capture the structure of high-performance or feasible regions in aerodynamic optimization problems. Secondly, LHS is an unconditional sampling method that lacks task awareness—it cannot generate samples conditioned on specific performance targets such as prescribed lift coefficient C_L or lift-to-drag ratio L/D . In data-driven optimization, however, it is often desirable to sample design points that

are more likely to yield optimal objective values on the surrogate model, a capability that LHS inherently lacks.

With this consideration, the quality of the data used to train the surrogate model becomes critical. An efficient geometry sampling method can help the design by enabling the exploration of diverse design alternatives. The general approaches are Design of Experiments (DoE) methods, which utilize statistical principles to sample within a search space in order to understand the relationship between input variables (factors) and the output response. Representative DoE methods include Latin Hypercube Sampling (LHS)[4], Uniform Design (UD)[5; 6], and Monte Carlo Design (MCD) [7; 8]. Although these methods are effective in achieving uniform coverage and statistical diversity, they typically do not account for the complex, nonlinear correlations or sparsity patterns that characterize high-performance regions in practical aerodynamic design spaces. Moreover, they are usually agnostic to the specific task objectives or physical constraints, resulting in significant portions of sampled data falling into infeasible or low-performance regions.

With specific constraint settings and nonlinear programming, some methods also achieve sampling with constraint handling, including probabilistic approaches (e.g., constrained Expected Improvement [EI] and Probability of Improvement [PI]) and classical methods (e.g., penalty function and sequential quadratic programming [SQP]) [9]. While these strategies improve feasibility by incorporating constraints, they often require repetitive optimization for each new sample and can be computationally inefficient, especially as problem dimensionality grows. Furthermore, such methods may still struggle to efficiently target high-performance or rare design solutions, particularly in high-dimensional or multifidelity design tasks.

Diffusion models offer several advantages over traditional sampling approaches. They excel in searching design spaces that may contain multimodal or high-dimensional

landscapes with numerous local minima or maxima. Moreover, diffusion models can effectively handle non-differentiable or discontinuous objective functions. Unlike traditional methods relying on derivatives, diffusion models—which are based on stochastic dynamics—do not require differentiability, making them highly versatile in handling a wider range of objective functions. Additionally, these models can incorporate prior knowledge, enabling customized and guided exploration. The diffusion model has found application as a sampler in various scenarios for addressing inverse-design problems. These include fields such as bio-mechanics [10], aerodynamic design [11], and chemical design [12]. These diverse applications demonstrate the versatility and effectiveness of the diffusion model in generating solutions for complex design challenges across different domains. The ability of capturing complex dependencies and generating realistic responses further expands the applicability of diffusion models to generate more natural conversations and physical visualizations, extending even to fields such as Chat Bot (e.g., ChatGPT ¹) and video generations (e.g., Sora ²).

This paper primarily focuses on the challenge of high-dimensional, high-fidelity inverse design, which imposes significant demands on generative models in terms of handling large-scale data dimensions and achieving a high degree of precision in data modeling. These requirements necessitate advanced techniques to effectively capture complex design relationships and accurately reflect intricate physical phenomena within the generative process. To address this challenges, from a geometric perspective, we employ a reduced-order modeling (ROM) strategy based on modal parameterization to globally represent wings. From a generative modeling perspec-

¹<https://chat.openai.com/>(last accessed on 14 June 2024).

²<https://openai.com/index/sora/>(last accessed on 14 June 2024).

tive, we use a conditional diffusion model based on an energy-based model (EBM). Unlike generative models such as GANs, this approach offers several advantages. Diffusion models are typically more stable during training, as they avoid the adversarial process that can lead to mode collapse in GANs, resulting in more reliable convergence. Additionally, diffusion models produce high-quality, diverse samples by progressively refining random noise into realistic designs, allowing for a more accurate representation of complex geometries. The conditional diffusion model also enables explicit conditioning, directly incorporating desired design constraints and making it more flexible in generating targeted designs that meet specific requirements. By leveraging an energy-based framework, this model can embed physical constraints and principles, leading to outputs that adhere more closely to underlying physics compared to GAN-based approaches. These advantages make conditional diffusion models a powerful and robust choice for high-dimensional, high-fidelity design applications.

This paper is organized as follows: In Section 2, we define the high-fidelity aerodynamic inverse design problem and outline the generative inverse design framework for aerodynamic applications. Section 3 presents the generative modeling approach, detailing the energy-based model (EBM), the energy-guided conditional diffusion model, and the value function that integrates inverse design requirements along with training procedures. In Section 4, we describe the dataset and experimental setup in detail. In Section 6, we summarize the key findings and contributions of this work.

2. Problem Description

In this section, we outline the problem, detailing the high-fidelity aerodynamic inverse design in Section 2.1. We then introduce the generative inverse design approach

in Section 2.2, highlighting its significance and distinguishing it from traditional discriminative methods.

2.1. High-fidelity inverse design problem

An inverse design problem can be mathematically described as finding a set of design parameters $\mathbf{x}$ that satisfies certain target performance metrics $\mathbf{y}_{\text{target}}$. In aerodynamic design, for example, this involves adjusting the shape of a geometry to achieve a desired aerodynamic performance (i.e. lowest drag, lift-to-drag ratio or pressure distributions). We can formulate an inverse design problem as:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathcal{L}(\mathbf{y}(\mathbf{x}), \mathbf{y}_{\text{target}}) \\ \text{subject to: } \quad & \mathcal{R}(f(\mathbf{x})) = 0, \quad (\text{governing equations, e.g., Navier-Stokes}) \\ & g(\mathbf{x}) \leq 0, \quad (\text{geometric or design constraints}) \\ & h(\mathbf{x}) \geq 0, \quad (\text{aerodynamic coefficient constraints}) \end{aligned} \quad (1)$$

where $\mathbf{x}$ is the vector of design parameters (i.e. geometric shape representation of a wing), $\mathbf{y}(\mathbf{x})$ is the performance metrics evaluated by a model, $\mathbf{y}_{\text{target}}$ is the desired or target performance. $(\mathbf{y}(\mathbf{x}), \mathbf{y}_{\text{target}})$ is a loss function that quantifies the difference between the predicted and target performance (e.g., mean squared error, L2 norm). The design process is constrained by physical laws, such as fluid dynamics equations, which govern how the design parameters affect the performance metrics. For aerodynamic problems, this might include the Navier-Stokes equations for fluid flow:

$$\mathcal{R}(g(\mathbf{x})) = 0 \quad (2)$$

where $g(\mathbf{x})$ is the flow field solution (e.g., velocity, pressure, density) that depends on the design parameters $\mathbf{x}$. $\mathcal{R}(g(\mathbf{x})) = 0$ represents the governing fluid dynamics

equations (e.g., Euler, Navier-Stokes equations) that must be satisfied for any design. Specifically in this study, we aim to explore the feasible design parameters $\mathbf{x}$ that satisfy the corresponding aerodynamic coefficient constraints.

2.2. Generative inverse design

The shift from discriminative to generative design in design thinking represents a transition from one-to-one solution mapping to a score-based evaluation of design outcomes. Traditional design thinking uses discriminative models. Discriminative models—such as logistic regression, support vector machines, random forests, and decision trees—are specifically designed to model the conditional probability of design condition $\mathbf{c}$ given by the design $\mathbf{x}$, follows

$$p(\mathbf{c} | \mathbf{x}). \quad (3)$$

Different from the discriminative model, generative models (i.e. Bayesian networks, Generative Adversarial Networks (GANs), etc) generate the design given by the condition $\mathbf{c}$, that generate new designs meeting a certain criteria of design conditions, following

$$p(\mathbf{x} | \mathbf{c}) = \frac{p(\mathbf{c} | \mathbf{x}) \cdot p(\mathbf{x})}{p(\mathbf{c})}, \quad (4)$$

based on Bayesian theorem.

The solution we use in this study for guiding the generative model is the energy-based model (EBM) [13], which is a probabilistic model that defines a probability distribution using an energy function, where lower energy states correspond to more probable configurations and higher energy states correspond to less probable ones. In this study, we leverage the theory of EBM that guide our diffusion model as an energy equation $\mathcal{E}(\cdot)$. The conditions integrated into this energy equation $\mathcal{E}(\cdot)$ are

implicit and typically involve hidden patterns and regularities present in the data. Furthermore, this conditional generative model can be represented by Equation 5.

$$p(\mathbf{x} \mid \mathbf{c}) \propto p(\mathbf{x}) \cdot e^{-\mathcal{E}(\mathbf{x})} = p(\mathbf{x}) \cdot e^{-\beta V(\mathbf{x})}, \quad (5)$$

where $\mathcal{E}(\mathbf{x})$ is an energy equation, $p(\mathbf{x})$ models the data distribution, $p(\mathbf{x} \mid \mathbf{c})$ specifies the geometry sample distribution under condition $\mathbf{c}$.

3. Methodology

In recent years, the diffusion models have demonstrated impressive performance in video generation, decision making and scientific simulations, etc. These models not only leverage their advantages in terms of model capacity but also possess the ability to generate new samples in high-dimensional space that adhere to physical laws, exhibit realistic effects, and diverge from the original dataset. Compared to other generative models, such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), etc, diffusion models trade off inference speed for better generation quality in modeling complex data distributions. In the meantime, to achieve comparable generation quality, diffusion models requires a smaller model architecture, leading to more efficient resource utilization and potentially reduced computational overhead, despite the slower inference time. This discussion is addressed in detail across eries of papers [14; 15; 16]. The fundamental principle can be traced back to the neural ordinary differential equations (Neural ODEs) [17], which discusses the advantages and disadvantages of generative modeling with continuous normalizing flows. The balance between quality and model size makes diffusion models particularly advantageous in scenarios where precision in generation is prioritized over speed.

3.1. Energy-guided conditional diffusion model

Diffusion models [18] are latent variable models expressed as $p_\theta(x_0) := \int p_\theta(x_{0:T}), dx_{1:T}$, where $x_1, \dots, x_T$ are latent variables with the same dimensionality as the observed data $x_0 \sim q(x_0)$. In this study, we utilize an unconditional diffusion model to model data distributions, coupled with an EBM that provides conditional guidance for the generative process, resulting in an energy-guided conditional diffusion model.

An unconditional diffusion model (shown in Figure 1) comprises a fixed forward process, which progressively adds Gaussian noise to the data, and a learned probabilistic reverse process, which removes the noise. The reverse process generates new samples from random noise, serving as the model's generative mechanism. Given a design prior dataset $\mathcal{D} = \{x_i\}_{i=1}^N \sim p_{\text{data}}(x)^{\otimes N}$, where p_{data} concentrates on a feasible set C , we will learn a generative model p such that $p \approx p_{\text{data}}$ using a DDPM. We first simulate a forward noising process starting from data distribution $p_0(x) = p_{\text{data}}(x)$ which converges to the standard Gaussian distribution $p_T(x) \approx \mathcal{N}(0, \text{Id})$ as $T \rightarrow \infty$. The forward process is defined by the following stochastic differential equation (SDE),

$$d\mathbf{x}_t = f(\mathbf{x}_t, t) dt + \sigma(t)dw_t, \quad \mathbf{x}_0 \sim p_{\text{data}}(x), \quad 0 \leq t \leq T, \quad (6)$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a vector-valued drift function, $\sigma(t): \mathbb{R} \rightarrow \mathbb{R}$ is a scalar-valued diffusion coefficient, and $(\mathbf{w}_t)_{t \geq 0}$ is a d -dimensional Brownian motion [19]. Next, with mild assumptions [20; 21] that are satisfied for the processes in this study, the reverse process that generates the data from normal noise admits a SDE description,

$$d\mathbf{x}_t = \left(f(\mathbf{x}_t, t) - \frac{\sigma^2(t)}{2} \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t, t) \right) dt, \quad (7)$$

where $-\frac{\sigma^2(t)}{2} \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t, t)$ is the score-based term that adjusts the drift by incorporating the score and gradient of the data probability, $\log p_\theta(\mathbf{x}_t, t)$, with respect to data $\mathbf{x}_t$. $\log p_\theta(\mathbf{x}_t, t)$ is computationally easier compared to directly model $p(x)$,

which has been discussed by Song *et al.* [22]. The score term encourages the reverse process to move toward regions with higher data density, transforming noise into structured data. The scaling factor $\frac{\sigma^2(t)}{2}$ adjusts the impact of the score based on the diffusion strength $\sigma(t)$.

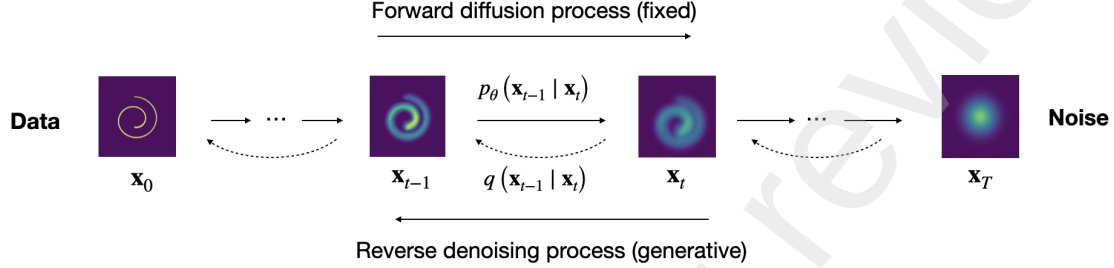


Figure 1: Forward diffusion process and reverse denoising process for a 2D swiss roll case.

Given by $p_{t|0}(\mathbf{x}_t | \mathbf{x}_0)$ as the conditional density of the forward SDE starting at $\mathbf{x}_t$, the guidance of conditional diffusion model is

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{c}) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \mathcal{E}_t(\mathbf{x}_t), \quad (8)$$

where diffusion model learns the data distribution $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$ and energy model provides the guidance $\nabla_{\mathbf{x}_t} \mathcal{E}_t(\mathbf{x}_t)$, not only the guidance defined by condition from the final state, but also evaluating the guidance during the intermediate state of diffusion model. We will introduce the details of learning this process in Section 3.3.

3.2. Value function assembling inverse design constraints

Leveraging the energy additivity principle in energy-based models (EBMs), the total energy can be expressed as the sum of independent components, each corresponding to different loss terms or constraint conditions:

$$p(x) = \frac{1}{\prod_{i=1}^n Z_i} \exp \left(- \sum_{i=1}^n \mathcal{E}_i(x) \right) = \prod_{i=1}^n p_i(x), \quad (9)$$

where

$$p(x) = \frac{1}{Z} e^{-\mathcal{E}(x)} = \frac{1}{Z} e^{-\beta V(x)}, \quad (10)$$

follows a Gibbs distribution [23]. The combination of two energy functions represents an "AND" relationship in terms of probabilities, meaning that the overall probability distribution is equal to the product of their probability densities. This product structure captures the joint behavior of the two components, increasing the model's flexibility and accuracy by incorporating complementary information from different sources or criteria. Additionally, this method can enhance generalization by combining distinct objectives without requiring them to be modeled separately. Follow this energy additivity of EBM, we can design our value function assembling inverse design constraints as:

$$V = \sum_{i=1}^n \lambda_i \cdot \mathcal{L}_i(\mathbf{y}(\mathbf{x}, \mathbf{c}), \mathbf{y}_{\text{target}}) + \sum_{j=1}^m \gamma_j \cdot R_j(\mathbf{x}, \mathbf{c}), \quad (11)$$

where $\mathbf{y}_{\text{target}}$ is the desired aerodynamic performance for the design $\mathbf{x}$ and $\mathbf{c}$ refers to the specific aerodynamic condition imposed on the design evaluations. $\mathcal{L}_i$ represents the i th loss function with λ_i weight controlling its contribution to the total loss and $R_j(\mathbf{x}, \mathbf{c})$ is the j th regularization term, used to control the complexity of the design parameters $\mathbf{x}$ to satisfy certain physical constraints. The specific design of value function for each case is shown in Section 4. The temperature coefficient β controls the generative model exploration on data, we will further investigate how its value affects the performance of generative model in Section 4.

3.3. Training

Instead of directly modeling $p(x)$, modeling score function

$$s_{\theta}(x_t) = \nabla_{x_t} \log p(x_t), \quad (12)$$

makes diffusion model computationally easier [22]. An unconditional diffusion model model the distribution of a given CFD aerodynamic dataset $\mathcal{D}_\mu$, referring to the theory from score-based generative modeling through stochastic differential equations [22], we involve Equation 13 as the training loss function,

$$\begin{aligned}\theta^* &= \arg \min_{\theta} \mathbb{E}_{(\mathbf{x}, \mathbf{c}) \sim \mathcal{D}_\mu, \epsilon, t} [\|\sigma_t \nabla_{\mathbf{c}_t} \log \mu_t(\mathbf{c}_t | \mathbf{x}) + \epsilon\|_2^2], \\ &= \arg \min_{\theta} \mathbb{E}_{(\mathbf{x}, \mathbf{c}) \sim \mathcal{D}_\mu, \epsilon, t} [\|\epsilon_\theta(\mathbf{c}_t | \mathbf{x}, t) - \epsilon\|_2^2],\end{aligned}\quad (13)$$

where θ^* represents the first process diffusion model parameters, $\epsilon_\theta(\cdot)$ represents the noise predicted by the model with parameters θ . Generally, diffusion model introduces the noise-weighted score function to compute the loss function in order to stabilize the training process. In our model, given diffusion input state s and time step t , the noise that the diffusion model with parameter θ predicts for the current action (or guidance) is $\epsilon_\theta(a_t | s, t)$, to scale this noise to the normal distribution $\epsilon \sim \mathcal{N}(0, I)$, we have

$$\epsilon_\theta(\mathbf{c}_t | \mathbf{x}, t) = -\sigma_t \nabla_{\mathbf{c}_t} \log \mu_t(\mathbf{c}_t | \mathbf{x}), \quad (14)$$

where $\mathbf{x}_t = \alpha_t \mathbf{c} + \sigma_t \epsilon$ follow the samples generated with condition and diffusion noise.

For the energy-guided conditional diffusion model, in addition to modeling the score function of the unconditional diffusion model, it is necessary to model an intermediate state energy function $\mathcal{E}_t(x_t)$. The gradient of this intermediate state energy function $\nabla_{x_t} \mathcal{E}_t(x_t)$, serves as energy guidance, which aids in calculating the score function for the conditional distribution shown in Equation 15:

$$\nabla_{x_t} \log p(x_t | c) = \nabla_{x_t} \log p(x_t) - \nabla_{x_t} \mathcal{E}_t(x_t) \quad (15)$$

Thus, $\nabla_{x_t} \log p(x_t)$ can be trained by denoising diffusion model [15] and we can use contrastive energy prediction (CEP) [24; 25] to acquire the energy guidance

$\nabla_{x_t} \mathcal{E}_t(x_t)$. Suppose we have a well-defined value function that is appropriate for investigating the problem at hand. CEP trains an intermediate state energy equation that is suitable for the current guidance policy. For a K step training process, the training objective is

$$\begin{aligned} \arg \min_{\phi} \mathcal{E}_{t,\mathbf{c},\epsilon^{1:K},\mathbf{x}^{1:K} \sim \mu(\mathbf{x}|\mathbf{c})} & \left[- \sum_{i=1}^K \frac{e^{\mathcal{E}(\mathbf{x},\mathbf{c})}}{\sum_{j=1}^K e^{V(\mathbf{x}^j,\mathbf{c})}} \log \frac{e^{(-f_{\phi}(\mathbf{x}_t^i,\mathbf{c},t))}}{\sum_{j=1}^K e^{-f_{\phi}(\mathbf{x}_t^j,\mathbf{c},t)}} \right], \\ \arg \min_{\phi} \mathcal{E}_{t,\mathbf{c},\epsilon^{1:K},\mathbf{x}^{1:K} \sim \mu(\mathbf{x}|\mathbf{c})} & \left[- \sum_{i=1}^K \frac{e^{V(\mathbf{x}^i,\mathbf{c})}}{\sum_{j=1}^K e^{V(\mathbf{x}^j,\mathbf{c})}} \log \frac{e^{(-f_{\phi}(\mathbf{x}_t^i,\mathbf{c},t))}}{\sum_{j=1}^K e^{-f_{\phi}(\mathbf{x}_t^j,\mathbf{c},t)}} \right], \end{aligned} \quad (16)$$

where the objective is to minimize a loss function with respect to ϕ that represents the parameters of a neural network model f_{ϕ} . t is the time variable indicates a step of diffusion model, $\epsilon^{1:K}$ are noise variables sampled in K times, $\mathbf{x}^{1:K}$ is a set of K design samples drawn from the conditional distribution $\mu(\mathbf{x}|\mathbf{c})$. After completing the training of the intermediate state energy function, we combine it with the value function as an optimal policy to guide the generation of diffusion model. The Algorithm 1 depicts the above whole process of achieving the value function guided conditional diffusion model designed for generating high aerodynamic performance wings.

Algorithm 1 Conditional diffusion Model for Generative Inverse Design

0: **Input:** Given CFD analyzed training wing dataset $\mathcal{D}_\mu$, which contains the design variable $\mathbf{x}$ and aerodynamic coefficients $\mathbf{y}$ information.

0: *//Training the diffusion model*

0: **for** each gradient step **do**:

0: Sample B data points $(\mathbf{x}, \mathbf{y})$ from $\mathcal{D}^\mu$, B Gaussian noises ϵ from $\mathcal{N}(0, \mathbf{I})$, and B time steps t from $\mathcal{U}(0, T)$

0: Perturb $\mathbf{x}$ according to $\mathbf{x}_t := \alpha_t \mathbf{x} + \sigma_t \epsilon$

0: Update $\theta \leftarrow \theta - \lambda_\theta \nabla_\theta \sum [\|\epsilon_\theta(\mathbf{x}_t | \mathbf{y}, t) - \epsilon\|_2^2]$

0: **end for**

0: *//Generating perturbed designs using the diffusion model*

0: **for** each state c in $\mathcal{D}^\mu$ **do**:

0: Sample K perturbed designs $\hat{\mathbf{x}}^{(1:K)}$ from the diffusion model $\mu_\theta(\cdot | c)$ and store them as $\mathcal{D}^{\mu_\theta}(c)$

0: **end for**

0: *//Training the evaluation model and the energy guidance model*

0: **for** each gradient step **do**:

0: Sample B data points $(c, \mathbf{a}, r, c')$ from $\mathcal{D}^\mu$, B Gaussian noises ϵ from $\mathcal{N}(0, \mathbf{I})$, and B time steps t from $\mathcal{U}(0, T)$

0: Calculate the target Value function $\mathcal{T}^\pi V_\psi(c, \mathbf{x})$ and detach gradient

0: Update $\psi \leftarrow \psi - \lambda_\psi \nabla_\psi \sum [\|V_\psi(c, \mathbf{x}) - \mathcal{T}^\pi V_\psi(c, \mathbf{x})\|_2^2]$

0: Perturb $\hat{\mathbf{x}}$ according to $\hat{\mathbf{x}}_t := \alpha_t \hat{\mathbf{x}} + \sigma_t \epsilon$

0: Update $\phi \leftarrow \phi + \lambda_\phi \nabla_\phi \sum_i \left[\frac{e^{\beta V_\psi(c, \hat{\mathbf{x}}_i)}}{\sum_j e^{\beta V_\psi(c, \hat{\mathbf{x}}_j)}} \log \frac{e^{f_\phi(\hat{\mathbf{x}}_{i,t} | c, t)}}{\sum_j e^{f_\phi(\hat{\mathbf{x}}_{j,t} | c, t)}} \right]$

0: **end for**

0: **Output:** Generated wing samples $\mathcal{D}_\mu = 0$

4. Experiments

In this section, we present the results and discussions of our study. The purpose of the experiment is to validate whether the proposed diffusion-based geometry sampling method can support the data-driven ASO, multi-fidelity design exploration, and further operation-aware design facing rapid industry design demand. To support this argument, we will discuss in the following three sections. Firstly, in Section 4.1, we will introduce our training data and design problem. Secondly, in Section 4.2, we do experiments using the conditional diffusion-based sampling method to test whether it can generate high-performance and high-fidelity wing samples. Finally, in Section 4.3, we demonstrate how diffusion-generated data can enhance surrogate-based aerodynamic shape optimization through several data-driven design cases.

4.1. Aerodynamic data and design problem

The wing geometry investigated in this paper is National Aeronautics and Space Administration (NASA) Common Research Model (CRM) wing configuration³, which is a contemporary supercritical transonic wing that is representative of a wide-body commercial transport aircraft. We adopt the compact modal parameterization method proposed by Li and Zhang [26] to represent the wing geometry. Specifically, we use the 50 wing shape modes extracted in their subsequent work [4], which capture the principal deformation patterns of the design space. We use ADflow [27]⁴ as CFD solver, which is a structured flow solver that solves the compressible Euler, laminar Navier-Stokes, and RANS equations using a second-order finite volume method, making it suitable for high-fidelity aerodynamic simulations. All data in

³<https://commonresearchmodel.larc.nasa.gov/home-2/high-speed-crm/experimental-approach/model-description/> (last accessed on March 8, 2026).

⁴<https://github.com/mdolab/adflow.git> (last accessed on March 8, 2026).

this paper follow the same data features format. The quantity of interests (QoIs) is $\mathbf{y} := \{C_L, C_D, C_M\}$. These QoIs are evaluated based on a given set of inputs $\mathbf{x} := \{\mathbf{x}_{\text{con}}, \mathbf{x}_{\text{geo}}\}$, which comprise operating condition information $\mathbf{x}_{\text{con}} := \{M, \alpha, h\}$ and geometry information $\mathbf{x}_{\text{geo}} := \{\alpha_{\text{twist}}, z\}$, where M is Mach number, α is angle of attack, h is flight altitude, α_{twist} is wing twist angle, and z are the coefficients to control mode combinations, which further control the wing shape perturbations. We evaluate the aerodynamic performance of mesh geometry at different mesh levels to obtain data of varying quality for subsequent experimental investigations. The mesh levels involve $L3$ (with a mesh size of 56,320) and $L2$ (with a mesh size of 450,560) meshes as shown in Figure 2.

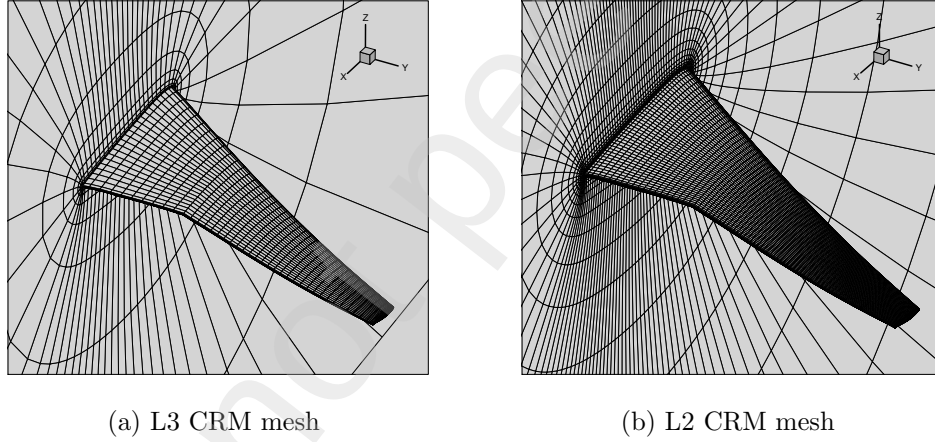


Figure 2: The low and high-fidelity CRM wing meshes used to generate CFD-based aerodynamic data in this work.

We aim to train the diffusion model using datasets comprising geometry and aerodynamic parameters. The goal is for the diffusion model to sample new data exhibiting similar characteristics. Subsequently, the sampled data will be remeshed

using `pyHyp`⁵ and subjected to Computational Fluid Dynamics (CFD) simulations to obtain aerodynamic estimations. We use `IDWarp`⁶ for mesh warping and `ADflow`⁷ as the CFD solver to analyze the perturbed wing meshes and obtain C_L , C_D , C_M evaluations. We will then compare the CFD-derived aerodynamic estimations with the aerodynamic estimations produced by the diffusion model. This comparison will serve to evaluate the reliability of the diffusion model in generating accurate aerodynamic estimations.

The design problem is a single-point aerodynamic shape optimization problem described as Table 1, following the guidance of the ADODG case 4.1⁸. The operating conditions of the wing are set to flight altitude 11,740 m and Mach number 0.85. We employ strict thickness constraints rather than a loose volume constraint to ensure practical design [28].

⁵<https://github.com/mdolab/pyhyp.git> (last accessed on March 8, 2026).

⁶<https://github.com/mdolab/idwarp.git> (last accessed on March 8, 2026).

⁷<https://github.com/mdolab/adflow/> (last accessed on March 8, 2026).

⁸<https://sites.google.com/view/mcgill-computational-aerogroup/adodg>(last accessed on 14 June 2024).

Table 1: ASO problem formulation as a drag minimization problem.

	Function/variable	Description	Quantity
Minimize	C_D	Drag coefficient	1
With respect to	α	Angle of attack	1
	α_{twist}	Angles of twists	7
	z	Wing shape modes perturbation	50
		Total design variables	58
Subject to	$C_L = 0.5$	Lift constraint	1
	$C_{M_y} \geq -0.17$	Pitching moment constraint	1
	$t \geq 0.98 \times t_{\text{initial}}$	Thickness constraints	750
	$V \leq 1.02 \times V_{\text{initial}}$	Maximum volume constraint	1
	$V \geq V_{\text{initial}}$	Minimum volume constraint	1
	$\Delta z_{\text{LE, upper, root}} = -\Delta z_{\text{LE, lower, root}}$	Fixed leading-edge constraints	8
	$\Delta z_{\text{TE, upper}} = -\Delta z_{\text{TE, lower}}$	Fixed trailing-edge constraints	8
		Total constraints	770

4.2. Conditional diffusion model for high-performance wing sampling

In this section, we present the results of wing data generation using a conditional diffusion-based sampling method. Our goal is to experimentally examine two key aspects. First, we assess whether the generated geometries satisfy the prescribed conditions. Second, we evaluate the extent to which the performance of the generated geometries, specifically their L/D ratios, degrades as the mesh fidelity increases. In other words, we investigate whether high- L/D designs generated under coarse fidelity levels remain effective at higher fidelities, and we quantify any observed performance drop. To address these questions, we conduct the following experiments.

We follow the conditional diffusion-based sampling procedure described in Sec-

tion 3, using the value function defined in Equation 11 to guide generation. The diffusion model is trained on 135,108 L3-fidelity CRM samples within the same design space and data feature format described in Section 4.1. After training, we generate new geometry samples conditioned on target performance metrics and evaluate their L/D ratios using the CFD pipeline. We validate with ADflow to assess whether generated samples align with the conditions.

For the optimization problem 1 investigating in this paper, the sampled wings need to satisfy that $C_L = 0.5$ and $C_M > -0.17$ (evaluated at Mach number 0.85 and flight altitude 11,740 m). The value function is then formulated as

$$\begin{aligned} V(x) = & \beta \times [-(C_{D_{\text{est}}} * 10 - 0.023)^2 \\ & - \lambda_1 (\max(\max(C_{L_{\text{est}}} - 0.505, 0), \max(0, 0.495 - C_{L_{\text{est}}}))^2 \\ & - \lambda_2 (\max(\max(C_{M_{\text{est}}} + 0.168, 0), \max(0, C_{M_{\text{est}}} + 0.170)))^2. \end{aligned} \quad (17)$$

It is worth noting that the condition $C_D \approx 0.0229$ is derived *a priori* through CFD-based optimization. From the consideration of experimental efficiency, we utilize this prior information as part of the value function information. To increase the variations in the sample data and prepare for higher-fidelity data in the sequential process, we allocate 1% margin for each coefficient in the value function. This is because high-fidelity aerodynamic coefficient evaluations differ significantly from low-fidelity evaluations, and we do not want to restrict the design space strictly during the initial sampling process. To ensure consistent thresholds for the variations of C_L , C_D , and C_M , we magnify the variations of the C_D value by a factor of 10. This adjustment allows us to align balanced thresholds of these parameters. When applying this approach to future optimization problems, the value function should be designed specifically based on their specific requirements.

Firstly, we compare the conditional diffusion-based sampling method with LHS,

as shown in Figure 3.

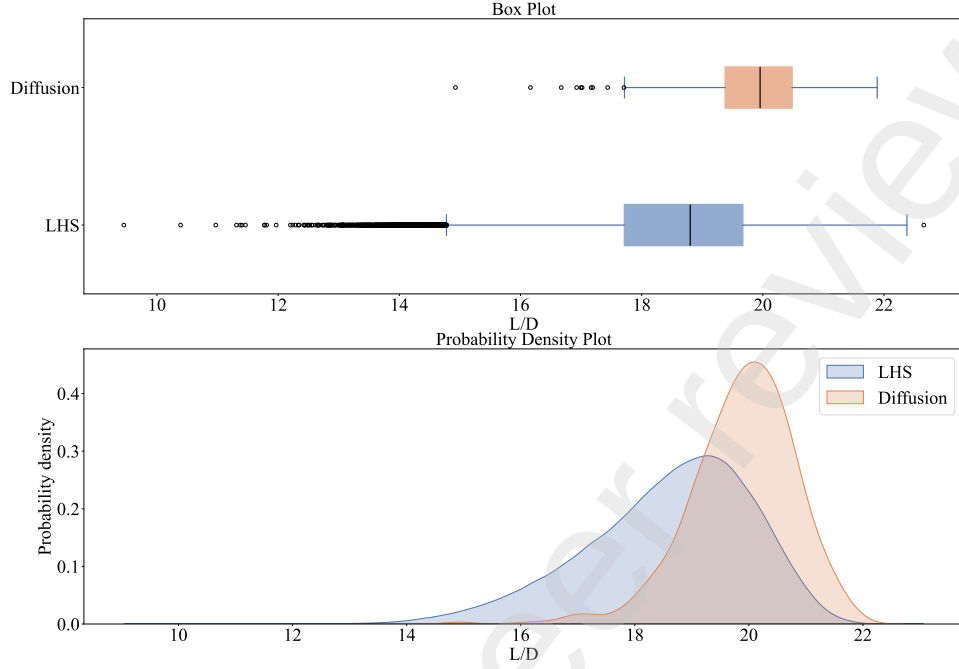


Figure 3: The box plot and probability density plot of conditional diffusion-based sampling method generates better aerodynamic performance (higher L/D) data than LHS method.

we can see that compared to the LHS-based method, the diffusion-based sampling generative data are more concentrated on higher L/D ratio area. Table 2 shows the generated wing samples evaluated L/D with respect to different β , more discussions on how β affects the sampling will be discussed in Section 5.1. Diffusion models add Gaussian noise to data through a forward process, transforming the data distribution into a tractable Gaussian prior. The reverse process then learns to denoise this Gaussian back into realistic samples. As a result, diffusion-based sampling inherently follows Gaussian distributions. In our conditional framework, increasing the temperature coefficient β sharpens the distribution of generated wing samples. In the limit as $\beta \rightarrow \infty$, this distribution is hypothesized to converge to the Gaussian

mean, effectively resembling a Dirac delta distribution [29].

Table 2: Generative wing samples with varying temperature coefficient β .

L/D threshold	$\beta = 8$	$\beta = 16$	$\beta = 32$	$\beta = 64$	$\beta = 128$
≤ 18	0.08	0.12	0.03	0	0.06
18–19	0.27	0.23	0.16	0.07	0.16
19–20	0.41	0.34	0.45	0.48	0.51
≥ 20	0.24	0.31	0.36	0.45	0.27

Next, we evaluate the ability of the conditional diffusion-based sampling method to generate multifidelity-consistent samples. Specifically, high-fidelity optimized wing shapes are expected to exhibit superior aerodynamic performance compared to those optimized under low-fidelity settings, due to their more accurate flow resolution and refined mesh representation. In this experiment, we condition the diffusion model—trained on low-fidelity (L3) data—using target L/D values derived from high-fidelity optimization. Our objective is to investigate whether the model can infer the underlying shape patterns required to meet high-fidelity aerodynamic targets, based solely on the trends learned from low-fidelity geometry–performance relationships. The sampling results are shown in Figure 4

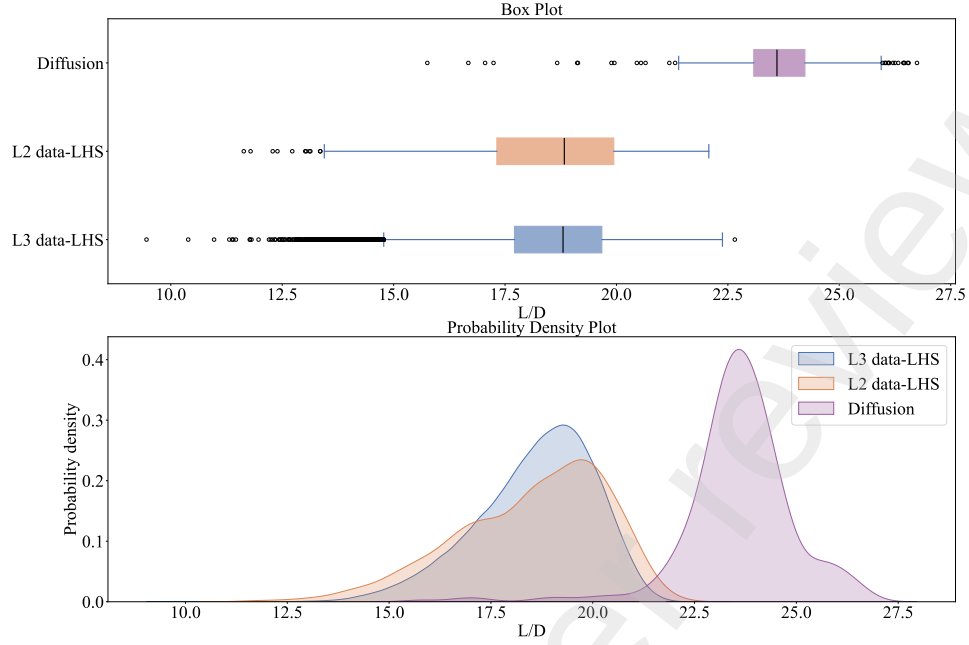


Figure 4: The box plot and probability density plot show that diffusion-based sampling method generates better aerodynamic performance (larger L/D) data validated on L2 mesh than directly LHS on L2 mesh data.

From Figure 4, we compare three different cases of sampling. The notations for each sampling case are as follows. "L3 Data - LHS": Geometry sampled using the LHS method and solved using CFD on the L3 mesh; "L2 Data - LHS": Geometry sampled using the LHS method and solved using CFD on the L2 mesh. "Diffusion": Geometry sampled using the conditional diffusion-based sampling method and solved using CFD on the L2 mesh. The data volume of each case is: "L3 Data - LHS": 140,000, "L2 Data - LHS": 2,000, "Diffusion": 500. From this figure, it is evident that the diffusion model is capable of generating wing samples with high L/D ratios that could not have been achieved by using the LHS method. This provides us with a greater number of high-quality design samples and design explorations for future

multi-fidelity design optimization.

4.3. Surrogate-based ASO

In this section, we present how conditional diffusion-based generative data support data-driven ASO. The procedures of data-driven ASO experiments are as follows: we build the surrogate model of our data-driven optimization framework based on the given training aerodynamic data. The trained surrogate model behaves as the aerodynamic coefficient estimator in the data-driven optimization framework. We use MLP as the basic structure of the surrogate model to perform data-driven ASO using only one fidelity level (specifically L3 level of CRM mesh in this experiment) of aerodynamic data. We train the surrogate models using data generated from different sampling methods, LHS and our conditional diffusion-based sampling method described in Section 4.2. These methods result in varying quantities of training data for training the surrogate models. The main objective of this experiment is to validate the impact of two different sampling methods on data-driven ASO. In order to ensure a fair comparison, we use the MLP as the basic framework for training, rather than other forms of neural networks. The specific configurations, such as the depth and number of nodes in the network, varies depending on the size of the training dataset. We use forward differentiation to estimate the gradient of the surrogate model when utilizing it for gradient-based optimization. This method serves as an approximation technique to compute the gradients of the objective function or constraints with respect to the input variables. Afterwards, we perform the data-driven ASO under a gradient-based optimization framework using a nonlinear programming optimizer SLSQP, which is integrated in `pyOptSparse`⁹. After com-

⁹<https://github.com/mdolab/pyoptsparse.git>(last accessed on 14 June 2024).

pleting the data-driven procedure, we proceed to analyze the optimized wing using a CFD solver for validation. We then compare the results of the data-driven optimization with those obtained using a conventional approach, which relies on CFD for estimating aerodynamic coefficients.

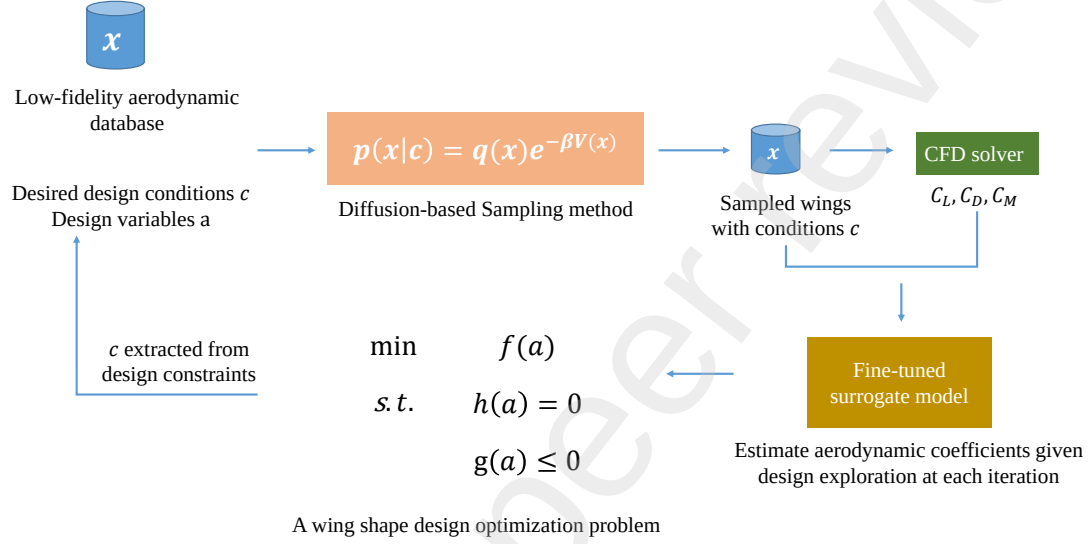


Figure 5: Work flow of data-driven wing shape design optimization with conditional diffusion-based geometry sampling.

We verify this data-driven optimization framework in a drag minimization problem summarized in Table 1, following the guidance of the ADODG case 4.1¹⁰. The operating conditions of wing are set to flight altitude 11,740 m and Mach number 0.85. We employ strict thickness constraints rather than a loose volume constraint to ensure practical design [28]. As stated in Section 4.1, we utilize a compact modal

¹⁰<https://sites.google.com/view/mcgill-computational-aerogroup/adodg>(last accessed on 14 June 2024).

parameterization method for wing geometry parameterization. Consequently, we employ wing shape mode coefficients to govern the perturbations in wing shape, which are also designated as the design variables. For the generated data with conditional diffusion model, we leverage fine-tuning techniques to modify the output layer of a trained surrogate model using 70,000 LHS dataset.

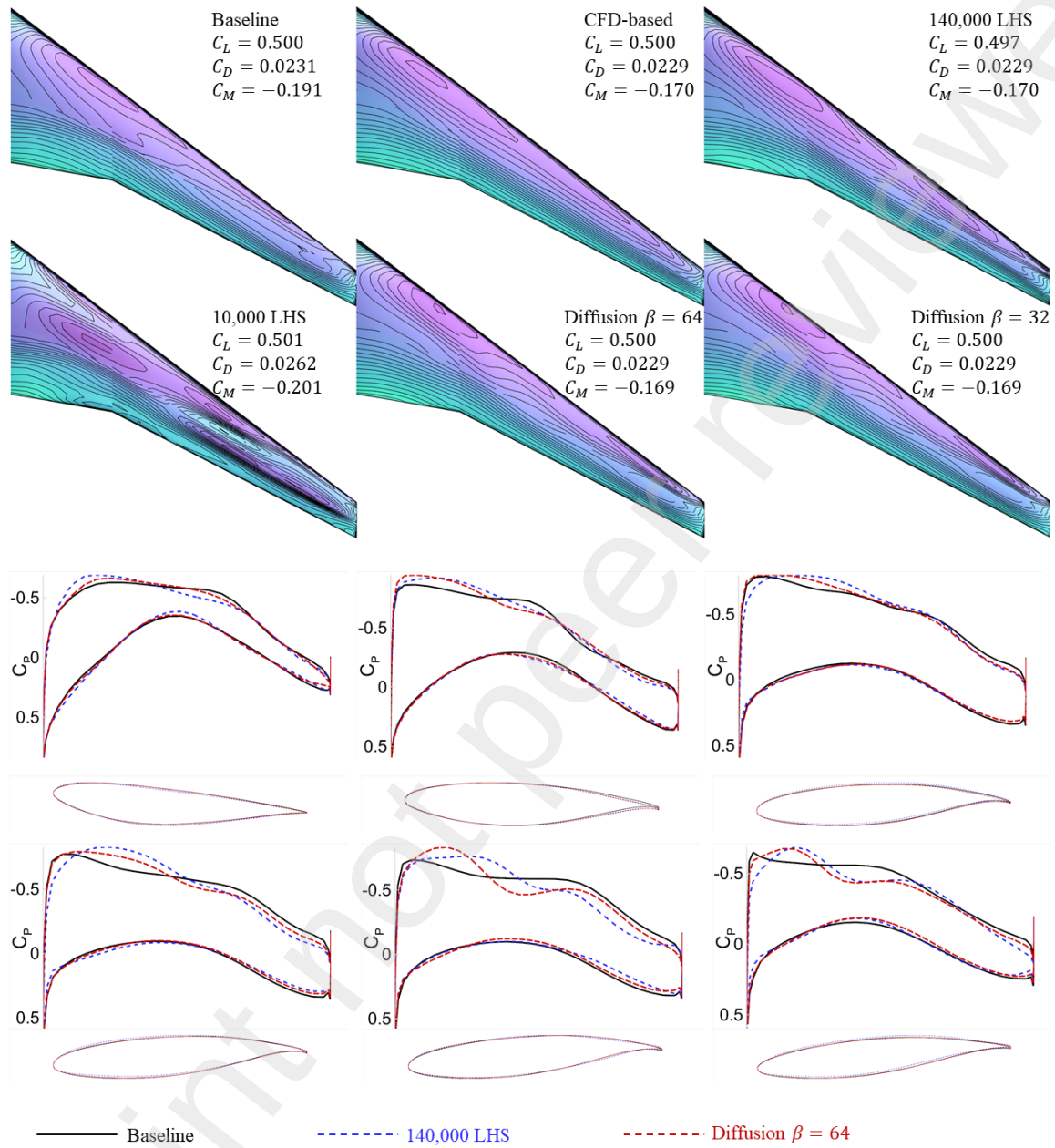


Figure 6: Wings and associated C_p contour of Baseline geometry, CFD-based and Data-driven optimized wings. Data-driven cases involve using 140,000, 10,000 LHS data and 500 Diffusion-based sampling Data when $\beta = 32$ and 64.

5. Discussions

5.1. *Exploration and exploitation of temperature coefficients*

In EBM, the temperature coefficient β fundamentally represents the balance between exploration and exploitation capabilities of the generative model with respect to data. Generally, as β increases, the model places greater emphasis on exploring the function $V(x)$, and we tend to rely more on the generative judgment of V . This balance allows the model to adapt dynamically between local search (exploitation) and broad search (exploration), impacting its performance in navigating complex design spaces. To examine this, we observe the distribution of sampled wing data for our quantity of interest (QoI), the lift-to-drag (L/D) ratio, under different β settings, as shown in Figure 7.

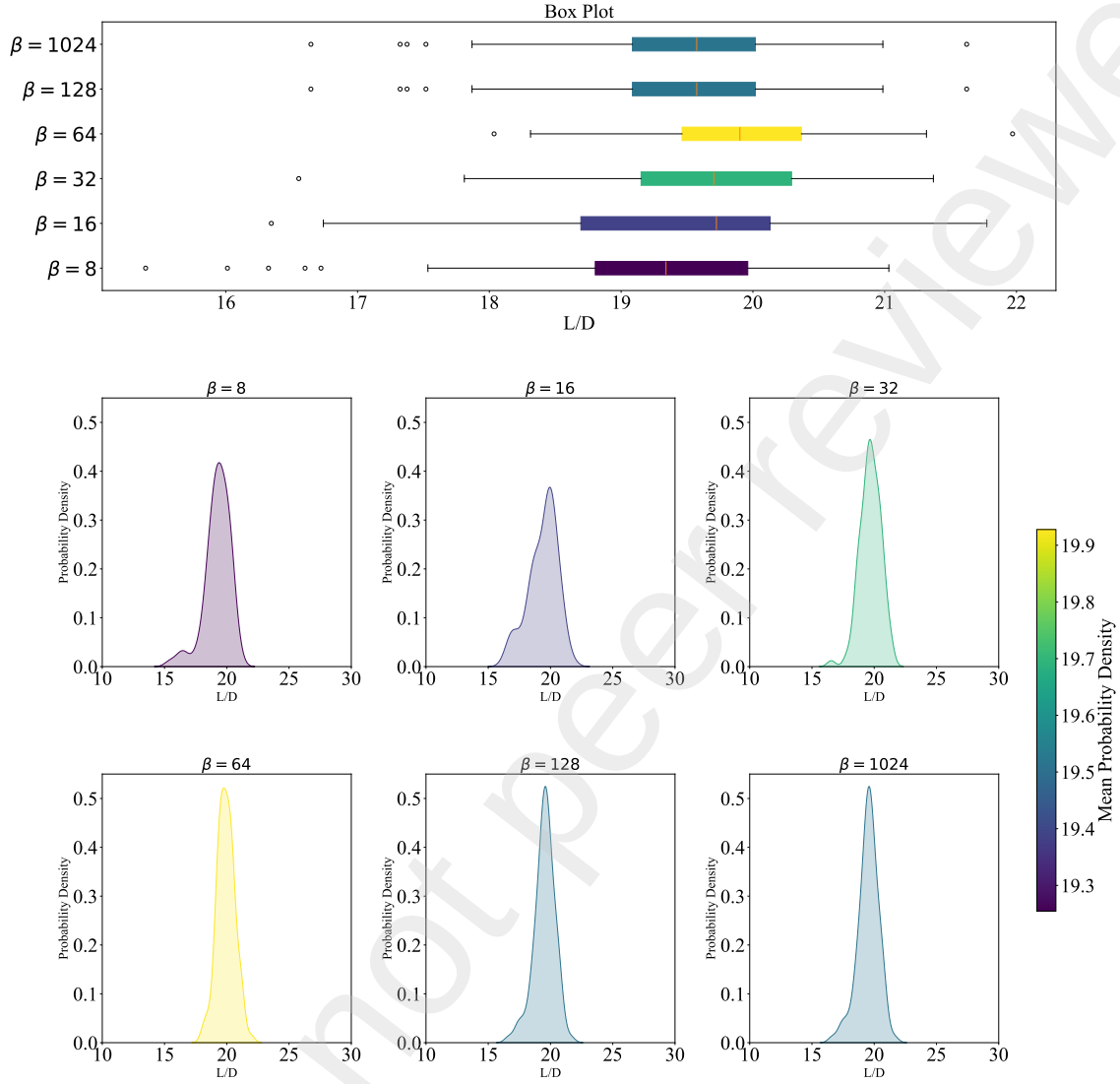


Figure 7: Data distributions generated through conditional diffusion-based sampling, varying by temperature coefficient β .

Our observations reveal that as β increases, the distribution of sampled wing L/D values becomes more concentrated, forming sharper peaks. This indicates that a higher β value enhances the model's ability to focus samples within the high QoI

region, effectively improving exploration across the design space. This capability is particularly advantageous in generative inverse design, as it allows the model to identify previously unexplored but potentially optimal designs. By carefully tuning β , the model achieves a dynamic balance that both leverages known high-performance regions and searches for unknown areas of potential.

5.2. Computational cost on surrogate-based ASO

The running results are summarized in Table 3. For the LHS, our investigations on training data are mainly on training data volume, data selection percentage; For the diffusion-based sampling method, our investigations on training data are on training data volume, temperature coefficient β and calculated L/D term in value function setting. We define a successful data-driven ASO as the optimization of the wing shape that, when evaluated using CFD solver, satisfies the constraint requirements of the optimization problem 1. We calculate the relative errors ε of aerodynamic coefficients estimations using trained surrogate models,

$$\varepsilon = |(y_{SM} - y_{CFD}) / y_{CFD}| \times 100\%, \quad (18)$$

where y_{SM} and y_{CFD} represent the estimation value of corresponding aerodynamic coefficients obtained using surrogate model and CFD simulations, respectively. From the experiments summarized in Table 3, we can observe that compared to the LHS method, diffusion-based method generates higher L/D ratio wing samples based on the given value function. The results obtained exclude the parts of wing samples that have low L/D ratio. While LHS offers a comprehensive search space for design, it also introduces a substantial number of low aerodynamic performance wing samples. This increases the computational burden for subsequent CFD calculations on these samples. In contrast, the conditional diffusion model generates wing samples guided

		Surrogate-based design				Generative design	
Training data	Training data volume	130,000	70,000	10,000	500	500	500
	Percentage (%) / β	100%	50%	7.6%	0.38%	32	64
	Value function (L/D)	-	-	-	-	21	20
	L/D bounds	9.45 – 22.66				15.56 – 22.6	18.17 – 22.06
Surrogate model	$\varepsilon(C_L)$	0.20	0.39	1.10	3.20	0.15	0.14
	$\varepsilon(C_D)$	0.35	0.62	1.94	5.44	0.24	0.22
	$\varepsilon(C_M)$	0.36	0.70	1.77	4.93	0.29	0.27
Optimization	Succeed	✓	-	-	-	✓	✓
	C_L	0.497	0.5	0.501	0.501	0.499	0.499
	C_D (counts)	229	267	261	279	229	229
	C_M	-0.17	-0.205	-0.2	-0.22	-0.17	-0.17
Computational cost	Core hours	56,000	28,000	4,000	200	200	200

Table 3: Data-driven ASO using LHS and diffusion-based sampling method with different size of training data. Resource utilization and design performance using CFD-based, surrogate-based, and generative approaches.

by constraints on parameters such as C_L , C_D , and C_M . As a result, the diffusion-based sampling method produces data with higher L/D ratios. Additionally, since the diffusion model is trained using aerodynamic data generated from CFD simulations, the sampling process aligns with this physics-based prior knowledge, ensuring the physical validity of the generated wing samples.

6. Conclusion

In this paper, we presented an effective conditional diffusion-based geometry sampling method that could generate high aerodynamic performance data with associated value function designed. Inspired by the energy conditional diffusion model theory, we proposed a value function-guided diffusion model to generate high performance wing samples to offer an efficient mean to yield high-fidelity wing design. We validated the effectiveness of this conditional diffusion-based sampling method in three key aspects, namely in generating desired high L/D ratio wing samples, supporting multi-fidelity design explorations, and enabling data-driven wing shape design optimization. The associated experiments have shown that with appropriate value function designed, the conditional diffusion model can generate desired L/D ratio wing samples with a normal distribution. Furthermore, the wing samples trained using low-fidelity data continue to adhere to this distribution when evaluated on a high-fidelity mesh, satisfying our requirements for the aerodynamic performance of wing samples under high-fidelity design. We fine-tuned our model to enable this generative data to support our data-driven aerodynamic shape optimization and get the results obtained are competitive with CFD-based design optimization.

Through experiment in Section 4.2, we show that with desired value function designed, we can generate the desired $L/D = 20$ wing samples using conditional diffusion model. We then validate that we can generate higher L/D ratio wing samples

from low-fidelity data that fits for a higher fidelity design. And with value function adjustment, we successfully generate the wing samples that follow a Gaussian distributions with a mean value around $L/D = 23$. In Section 4.3, we successfully use the generated wing samples to fine-tuning a preliminary surrogate model trained with LHS data. By adopting this approach, we can reduce the demand for the training set by 50.5%, while also decreasing the required CFD computations. It is worth noting that the establishment of this training set is aimed at addressing a series of optimization designs, as well as meeting the rapid high-fidelity design requirements for future mission-based and operation-aware aircraft designs.

In future work, we will investigate more geometry constraints imposed on the diffusion model (including wing span area, wing thickness, aspect ratio, etc.) to support more comprehensive design explorations. In the meantime, we aim to extend the exploration of data not only limited to high-fidelity design but also to include a broader range of wing structures for various types of aircraft and flight missions. This expansion will allow us to tackle design tasks that cater to different aircraft and flight mission requirements, thereby supporting an efficient operation-aware aircraft design optimization.

7. Acknowledgment

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Appendix A. Compact modal parameterization

The compact modal parameterization approach is a reduced-order representation of wing shape geometry, achieved by performing singular value decomposition (SVD)

on the wing sectional airfoils. The SVD for model order reduction is described briefly below. Each wing geometry sample is represented by $n = 8 \times 251$ section points, where there are 251 points for each of the eight sectional airfoils. Suppose we have m wing samples, we can assemble the snapshot matrix as

$$\mathbf{G} = \begin{bmatrix} z_1^1 - z_1^{\text{CRM}} & z_1^2 - z_1^{\text{CRM}} & \cdots & z_1^m - z_1^{\text{CRM}} \\ z_2^1 - z_2^{\text{CRM}} & z_2^2 - z_2^{\text{CRM}} & \cdots & z_2^m - z_2^{\text{CRM}} \\ \vdots & \vdots & \ddots & \vdots \\ z_n^1 - z_n^{\text{CRM}} & z_n^2 - z_n^{\text{CRM}} & \cdots & z_n^m - z_n^{\text{CRM}} \end{bmatrix}, \quad (\text{A.1})$$

where z_i^{CRM} is the i -th z -coordinate of the CRM baseline wing. In this study, we use 1,000 snapshots, i.e., $m = 1,000$. We then perform SVD on $\mathbf{G}$,

$$\mathbf{G} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T, \quad (\text{A.2})$$

where $\mathbf{U}$ is an $m \times m$ matrix where each column represents a wing mode shape, $\mathbf{\Sigma}$ is an $m \times n$ rectangular diagonal matrix with singular value entries in descending order, and $\mathbf{V}$ is the transpose of $\mathbf{U}$. Using all vectors in $\mathbf{U}$ as basis vectors, the wing geometry shape can be reconstructed via a linear combination operation with $\boldsymbol{\lambda}_{\text{full}} \in \mathbb{R}^m$ as a vector of coefficients. In other words, the coefficients $\boldsymbol{\lambda}$ can be used to define a new wing shape perturbation.

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